

Research Progress Report

BLG vs. β -Casein Adsorption at the Air–Water Interface

Prepared for: Prapasiri Pongprayoon (P.P.), COMFHA, Kasetsart University

Prepared by: Chalakon Pornjariyawatch

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1. Executive Summary

- **BLG is complete and validated:** 4.00 μ s of unbiased MD (4×1000 ns), fully PBC-corrected. The originally hypothesised two-factor adsorption gate is disproven (Pearson $r = +0.006$, 95% CI $[-0.09, +0.11]$); the paper is reframed as a precise scope claim.
- **CAS (β -casein) CENTER validation is now complete:** all 7 planned CENTER-only analyses finished (density, surface tension, DSSP, contact, N-terminal SASA, R_g , hydrogen bonds). This is the first full CAS analysis pass in the project.
- **CAS R1 (second replica) production is in progress on the cluster**, close to the expected 1000 ns finish based on CENTER’s observed rate — all CAS numbers below are CENTER-only ($n=1$) until R1 lands.
- Five questions below need your input before the comparative figures/table can be finalised.

2. BLG Results — Complete, 4 Independent Replicas

The central result: across 8006 analysed frames (CENTER + 3 replicas), 613 protein–interface contact events are detected, of which only 6 last ≥ 10 ns, and none progress to commitment. SASA ($24\text{--}37$ nm², PBC-corrected) and orientation angle θ show no correlation ($r = +0.006$) — ruling out $|r| > 0.11$. The protein remains globally compact ($R_g = 1.504 \pm 0.022$ nm, CENTER) and its hydrophobic calyx stays accessible throughout (calyx SASA 3.81 ± 0.46 nm²) across every independent replica.

BLG quantity	CENTER	Range across 4 replicas
Calyx SASA (nm ²)	3.81 ± 0.46	3.15 – 3.81 (mean)
R_g (nm)	1.504 ± 0.022	1.493 – 1.504 (mean)
Patch RMSD (nm)	0.198	0.198 – 0.347 (R1 elevated, see note)
DSSP Helix/Sheet/Coil (%)	11.8 / 36.7 / 51.5	consistent within ± 2 pp all 4
Surface tension (mN/m)	51.9 ± 38.5	51.9 – 52.7 (mean), matches TIP3P lit.

Note on R1: R1’s calyx-patch RMSD (0.347 nm) is elevated vs. the other 3 replicas (0.20–0.28 nm) — traced to R1 having 4 of the project’s 6 total long contact events; this is a real, contact-correlated conformational response, not noise, and should not be averaged into a flat cross-replica mean in the comparative figure.

3. CAS (β -Casein) Results — CENTER Complete (7/7), R1 In Progress

All numbers below are from the CENTER trajectory only ($n=1$); treat as provisional until the second replica (R1) lands. Read live from `results/analysis/cas*.npz` on the date of this report.

CAS quantity (CENTER)	Value
N-terminal (res. 1–25) SASA (nm ²)	25.41 ± 1.97
R _g (nm)	2.652 ± 0.209
DSSP Helix/Sheet/Coil (%)	25.2 / 3.8 / 71.0
Surface tension (mN/m)	51.4 ± 34.2
H-bonds: protein–protein	97.8 ± 9.7
H-bonds: protein–water	565.5 ± 25.1
H-bonds: interface water–water	12750.5 ± 1014

Interpretation: CAS’s coil-dominant secondary structure (71.0%) and much larger, more variable R_g (2.65 nm, std 0.21 nm vs. BLG’s 1.50 nm, std 0.02 nm) are consistent with its expected intrinsically-disordered character. Surface tension (51.4 mN/m) closely matches both BLG’s own value and the TIP3P literature (~50–52 mN/m) — a good physical cross-check that the CAS system setup is behaving correctly. CAS’s higher protein–water H-bond count (565.5 vs. BLG’s 393.2) is consistent with an IDP presenting more exposed backbone to solvent, matching the pre-registered prediction in the analysis script’s own documentation. **The headline N-term SASA (25.41 nm², 25 residues) vs. BLG’s calyx SASA (3.81 nm², 9 residues) is not yet a fair comparison** — see Question 1 below.

4. Questions for P.P.

1. **SASA normalisation.** CAS’s N-terminal SASA covers 25 residues; BLG’s calyx covers 9. A raw comparison is apples-to-oranges. Should this be normalised per-residue, or reported as relative SASA (% of a fully extended reference), before it becomes the paper’s headline comparative metric?
2. **“Secondary SASA”** — your June 9 note listed this separately from the DSSP table row. Our reading: SASA computed per secondary-structure element (DSSP assigns the region, SASA partitions across it). Is that the intended meaning?
3. **Which lab experiments to correlate against?** We assume this means the published experimental literature already surveyed (Cornec1999, Ulaganathan2017a for BLG; Mackie1999, Atkinson1995 for CAS) rather than in-house wet-lab data — please confirm, or point us to a specific dataset if one exists.
4. **How prescriptive should the applied “modify to adsorb” claim be?** Current data supports a comparative design-principle statement (disorder + exposed hydrophobic patches lower the kinetic barrier) but not an explicit mutation-level prescription. Is the softer framing sufficient?
5. **IFSC2026 poster (Sept 25 abstract deadline, 21 days from this report’s date)** — is this still happening? A July 16 report to you recommended leading BLG-only with CAS as an explicit “in progress” preview; we have no record of your reply. Please confirm either way so we can plan the next 3 weeks accordingly.

All numbers above are PBC-corrected and read live from `results/analysis/*.npz` as of this report’s generation date (2026-09-04) — not to be treated as frozen if analysis reruns later. Full open-item detail and history: `docs/PP_FEEDBACK_LOG.md`.